🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD MPP.4

Math is Ingenuity

Lesson 10-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratios and models to explain how the Penny Farthing and gear-driven bicycles were ingenious solutions to a real problem.

Language Objective: I can explain a gear ratio using the words rotation, teeth, and 'for each ____, the wheel turns ____.'

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

ingenuity | rotation | gear | ratio | equivalent ratios

👆 *Tap any word to see what it means and a picture.*

1. ________ — Coming up with different, and sometimes unique, solutions to problems.

2. ________ — One full turn of a wheel, gear, or pedal all the way around.

3. ________ — A toothed wheel that transfers motion through a chain or another gear.

4. ________ — A comparison of two quantities, like pedal rotations to wheel rotations.

5. ________ — Ratios that make the same comparison using different numbers.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How will your future vehicle be powered, and what math will be used in its design or use?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **ingenuity** — Coming up with different, and sometimes unique, solutions to problems.
  ingenio — Idear soluciones diferentes, y a veces únicas, a los problemas.
- [ ] **rotation** — One full turn of a wheel, gear, or pedal all the way around.
  rotación — Una vuelta completa de una rueda, un engranaje o un pedal.
- [ ] **gear** — A toothed wheel that transfers motion through a chain or another gear.
  engranaje — Una rueda dentada que transfiere movimiento mediante una cadena u otro engranaje.
- [ ] **ratio** — A comparison of two quantities, like pedal rotations to wheel rotations.
  razón — Una comparación de dos cantidades, como rotaciones del pedal y rotaciones de la rueda.
- [ ] **equivalent ratios** — Ratios that make the same comparison using different numbers.
  razones equivalentes — Razones que hacen la misma comparación con números diferentes.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

Describe a future mode of individual transportation. A complete answer says how it is ____, names a ____ (or rate) that shapes it, and explains where that math appears in the vehicle's ____.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: One pedal rotation makes one wheel rotation, so a bigger wheel's longer perimeter carries the bicycle farther with each pedal push.

Evidence: A strong answer connects the 1-to-1 ratio to the wheel's perimeter, explaining a bigger wheel travels farther per pedal push. A weak answer just says 'bigger is faster' without naming the ratio or perimeter.

Reasoning: This evidence connects the definition of ingenuity to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Describe a future mode of individual transportation. A complete answer says how it is ____, names a ____ (or rate) that shapes it, and explains where that math appears in the vehicle's ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** ingenuity
2. **Notes 2:** rotation
3. **Notes 3:** gear
4. **Notes 4:** ratio
5. **Notes 5:** equivalent ratios
6. **Watch for this mistake:** *Thinking a rear gear with MORE teeth makes the wheel spin more — it is the opposite: the 32-tooth gear gives only 1.5 rotations per pedal rotation, while the 12-tooth gear gives 4.*
7. **Try It 1:**
8. **Try It 2:**